Department of Computer Science (A.Y. 2025-2026)
Master of Computer Application (MCA)
&
Master of Science(Information Technology)
Semester - II
Subject: practical based on databases - beginner to professional (101300217)

Remarks:

- You might want to create a new folder with your **EnrollmentNo./YourName** for each exercise to keep them organized.

Exercise:1	**Basic SQL Syntax:**
1.	Create a database named university.
2.	Switch to the university database.
3.	Create a table named students with the following columns. Id(integer,primary key) Name(varchar(100),not null) Age(Integer,check age>16) Email(varchar(255),unique)
4.	Retrive the structure of the students table.
5	Insert at least 20 records.
Exercise:2	**Working with different data types:**
1.	Create a table named product with columns: Product_id(integer,primary key) Name(varchar(100)not null) Price(numeric(10,2),check price>0) Create_at(timestamp default current_timestamp) Is_avilable(Boolean default true)
2.	Insert sample data into products(20 data)
3.	Retrive all products where is_availble is true
4.	Update the price of a specific product.
5.	Delete a product based on product_id.
Exercise:3	**Modifying Tables and Constraints**
1.	Alter the students table to add a new column

	phone_number(numeric(11)unique)
2.	Rename the students table to university_students.
3.	Drop the email column from university_students.
4.	Add a not null constraint to the phone_number columns.
Exercise :4	**Inserting,Updating and Deleting Data.**
1.	Insert three student into university_students.
2.	Update the age of a student where id=1.
3.	Delete a student where id=3.
4.	Return data from modified rows.
5.	Insert multiple students in a single query.
6.	Remove all students who are younger than 18.
Exercise:5	**Retrieving Data**
1	Select all columns from university_students.
2	Select only name and age where age > 20.
3	Order results by name in descending order.
4	Limit the number of results to 2.
5	Select students whose names start with 'A'.
6	Retrieve distinct ages from the university_students table.
7	Find students who have NULL values in the phone_number column.
Exercise:6	**Using Operators and Functions**
1	Create a new table orders with columns: order_id (int, primary key) student_id (integer, foreign key referencing university_students(id)) total_price (numeric(10,2)) order_date (timestamp default current_timestamp)
2	Retrieve all students along with their orders using a JOIN.
3	Find the total orders placed by each student.
4	Retrieve students who have not placed any orders.
5	Retrieve students who have placed orders with a total price above 500.
6	Use a WITH query to find the average order price for each

	student.
Project:1	**Mini Project:** Library Management System using PostgreSQL
	Objective: Develop a system to manage books, students, and borrowing records. Step:1:DataBase Setup 1. Create a database named 'library_db'. 2. Switch to the database. Step 2: Creating Tables 1. Create a 'students' table with fields: student_id (primary key), name, age (must be >= 16), and phone_number (unique, not null). 2. Create a 'books' table with fields: book_id (primary key), title, author, and availability status. 3. Create a 'borrow_records' table with fields: record_id (primary key), student_id (foreign key), book_id (foreign key), borrow_date, and return_date. Step 3: Insert Sample Data 1. Insert at least 20 students records into the 'students' table. 2. Insert at least 10 books records into the 'books' table. 3. Insert at least 10 borrowing records into the 'borrow_records' table. Step 4: Queries and Data Retrieval 1. Retrieve the details of students who have borrowed books. 2. Find all available books. 3. Update the availability status of a book after it is borrowed. 4. Identify students who have not borrowed any books. 5. Record the return of a book and update availability. 6. Display first 5 students only. 7. Display next 5 students and skip first 5 8. Show only students who borrowed more than 1 book. 9. Find students who borrowed at least 3 books.

Project- 2	Employee Attendance Management System using PostgreSQL
	Step 1: Database Setup 1. Create a database named 'attendance_db'. 2. Switch to the database. Step 2: Creating Tables 1. Create an 'employees' table with fields: employee_id (primary key), name, department, and phone_number (unique, not null). 2. Create an 'attendance' table with fields: record_id (primary key), employee_id (foreign key), date, check_in_time, and check_out_time. Step 3: Insert Sample Data 1. Insert at least 20 employees records into the 'employees' table. 2. Insert at least 20 attendance records records into the 'attendance' table. Step 4: Queries and Data Retrieval 1. Retrieve the attendance records for a specific employee. 2. Find employees who have not checked in today. 3. Update the check-out time for a specific employee. 4. Identify employees who were present for more than 8 hours. 5. Retrieve a summary of total working hours for each employee.